

Inverse Design of Light-Trapping Layers using machine learning

Master Thesis



Abstract

This report details the surrogate optimization and active learning approaches used to inversely design light-trapping layers.

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1 Introduction

Introduce the problem of light-trapping layers here Author and Writer 2026.

2 Methods and theory

2.1 Thin Film Solar Cells

2.2 Rigorous Coupled Wave Analysis

This subsection will introduce the concept of Rigorous Coupled Wave Analysis (RCWA), a method in numerical photonics suited well for stacked periodic structures in the x and y axis with a period Λ_x, Λ_y respectively, and where each layer is homogeneous in the z -direction Kim and Lee 2023. The structure's optical properties are completely described by the relative permittivity ϵ and relative permeability μ of each layer, meaning that each structure will have a geometry described by

$$\epsilon(x, y, z)_N = \epsilon(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z} \quad (2.1)$$

and

$$\mu(x, y, z)_N = \mu(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z}, \quad (2.2)$$

where m and n are integers and the subscript N denotes the N 'th layer of the structure. For the entirety of this project, the designed fourier patterned light trapping layers will be designed using crystalline silicon (Si), Titanium Dioxide (TiO_2) and Silicon Nitride (Si_3N_4), which are all nonmagnetic, so the relative permeability $\mu = 1$. In general, ϵ and μ will be 3×3 tensors, but for isotropic and linear materials these are scalars. The equations solved for are of course the Maxwell equations, namely Faraday's law

$$\nabla \times \mathbf{E} = -\mu_0 \mu(x, y) \frac{\partial \mathbf{H}}{\partial t} \quad (2.3)$$

and Ampere's law

$$\nabla \times \mathbf{H} = \epsilon_0 \epsilon(x, y) \frac{\partial \mathbf{E}}{\partial t}, \quad (2.4)$$

Where ϵ_0 and μ_0 are the vacuum permittivity and vacuum permeability respectively. The structures are assumed free of charge and current,

$$\nabla \cdot (\epsilon \mathbf{E}) = 0$$

and

$$\nabla \cdot (\mu \mathbf{H}) = 0.$$

At a single frequency and in homogeneous media, the eigenmodes of Maxwell's equations are solved by plane waves, see Griffiths 2017 section 9.2.1 for example, meaning

$$\mathbf{E} = \mathbf{E}_0 \exp i(\mathbf{k} \cdot \mathbf{r} - \omega t) \quad (2.5)$$

and

$$\mathbf{H} = \mathbf{H}_0 \exp i(\mathbf{k} \cdot \mathbf{r} - \omega t), \quad (2.6)$$

Where $\mathbf{E}_0$ and $\mathbf{H}_0$ are the (complex) amplitudes of the fields, with the physical fields being the real parts of $\mathbf{E}$ and $\mathbf{H}$, $\mathbf{k}$ being the wavevector, i.e. the spatial frequency, and ω being the frequency of the fields, also given by the dispersion relation $\omega = \frac{c|\mathbf{k}|}{n(\lambda)}$, where λ is the wavelength of the light. It is noted that conservation of momentum leads to the spatial components of the wavevector only having two degrees of freedom, i.e.

$$|\mathbf{k}| = \frac{2\pi n(\lambda)}{\lambda} = \sqrt{k_x^2 + k_y^2 + k_z^2} = k_0, \quad (2.7)$$

and in general, for an incident angle θ and azimuthal angle ϕ , the x and y components will be chosen by

$$(k_x, k_y) = k_0(\sin \theta \cos \phi, \sin \theta \sin \phi). \quad (2.8)$$

Inserting equations 2.5 and 2.6 in equations 2.3 and 2.4 converts the time derivative to an algebraic factor of $-i\omega t$, meaning that the equations appear in their time harmonic form, common to many frequency domain numerical methods in photonics,

$$\nabla \times \mathbf{E} = i\omega\mu_0\mu(x, y)\mathbf{H} \quad (2.9)$$

$$\nabla \times \mathbf{H} = -i\omega\epsilon_0\epsilon(x, y)\mathbf{E}. \quad (2.10)$$

Owing to the periodicity of the system, see equations 2.1 and 2.2, both the electromagnetic fields and the permittivity and permeability may be written as their truncated fourier series in the x- and y-directions as below. Truncating the electromagnetic fields to order M, N in x, y makes RCWA a semi-analytical method precise only to this fourier order. The final fields are reconstructed as an inverse Fast Fourier Transform, so including enough fourier orders is an important convergence criterion.

$$\epsilon(x, y) = \sum_{m=-2M}^{2M} \sum_{n=-2N}^{2N} \epsilon_{m,n} \exp i(mL_x x + nL_y y), \quad (2.11)$$

$$\mu(x, y) = \sum_{m=-2M}^{2M} \sum_{n=-2N}^{2N} \mu_{m,n} \exp i(mL_x x + nL_y y), \quad (2.12)$$

$$\mathbf{E} = \exp i\mathbf{k} \cdot \mathbf{r} \sum_{m=-M}^M \sum_{n=-N}^N \mathbf{E}_{m,n} \exp i(mL_x x + nL_y y), \quad (2.13)$$

$$\mathbf{H} = \exp i\mathbf{k} \cdot \mathbf{r} \sum_{m=-M}^M \sum_{n=-N}^N \mathbf{H}_{m,n} \exp i(mL_x x + nL_y y), \quad (2.14)$$

where $L_x = \frac{2\pi}{\Lambda_x}$ and $L_y = \frac{2\pi}{\Lambda_y}$, and the parameters $\epsilon_{m,n}, \mu_{m,n}, \mathbf{E}_{m,n}$ and $\mathbf{H}_{m,n}$ are the fourier amplitudes of the m, n 'th diffraction order. It is worth mentioning, that since the momentum of light is always conserved, k_z implicitly depends on the expansion orders m, n , and writing out the k-vector in full, including the fourier components, yields

$$\mathbf{k}_{m,n} = (k_x + mL_x, k_y + nL_y, k_z(m, n)) = \quad (2.15)$$

$$= \left(k_x + mL_x, k_y + nL_y, \sqrt{k_0^2 - (k_x + mL_x)^2 - (k_y + nL_y)^2} \right), \quad (2.16)$$

where equation 2.7 has been used. The physical interpretation of the fourier series of the electromagnetic fields is then made obvious; the different terms are different diffraction orders, each with part of the initial z-component momentum kicked into either the x- or y-direction. It is noted how the permittivity and permeability is expanded to twice the order of the electromagnetic fields E and H . This arises from the fact that the algebraic components of equations 2.3 and 2.4 include products of Fourier series, so to reach order M, N in the electromagnetic fields, the permittivity and permeability must be truncated at twice the order. An example for just one fourier order will be given below, but for more details, I refer to Hwi Kim 2012. By inserting

2.3 Finite Element Method

2.4 Machine Learning Surrogates

2.5 Optimization Strategies

3 Results

Present the performance of the surrogate models and the final inverse design outcomes.

Figure 3.1: Placeholder for a results figure.

4 Discussion

Discuss the results and what worked/didn't work.

5 Outlook

Discuss future work.

6 Conclusion

Summarize the findings.

Bibliography

- Author, A. and B. Writer (2026). "A study on light-trapping nanostructures". In: *Journal of Nanophotonics* 10, pp. 1–10.
- Kim, Changhyun and Byoungcho Lee (2023). "TORCWA: GPU-accelerated Fourier modal method and gradient-based optimization for metasurface design". In: *Computer Physics Communications* 282, p. 108552. ISSN: 0010-4655. DOI: <https://doi.org/10.1016/j.cpc.2022.108552>. URL: <https://www.sciencedirect.com/science/article/pii/S0010465522002715>.
- Griffiths, David J. (2017). *Introduction to Electrodynamics, fourth edition*. Cambridge University Press.
- Hwi Kim, Junghyun Park Byoungcho Lee (2012). *Fourier Modal Method and Its Applications in Computational Nanophotonics*. CRC press.